SS-C-185a
NOVEMBER 21, 1960
SUPERSEDING
Fed. Spec. SS-C-185
April 19, 1954

FEDERAL SPECIFICATION

CEMENT, NATURAL, (FOR USE AS A BLEND WITH PORTLAND CEMENT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 **Types.**—The natural cement covered by this specification shall be furnished in the following types as specified. When the type of cement is not specified, type NA shall be furnished.

Type N. — Natural cement for use with portland cement in general concrete construction.

Type NA. — Natural cement, air entraining; for use with portland cement in general concrete construction.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, forms a part of this specification:

Federal Specifications:

SS-C-192 — Cement, Portland.
UU-S-48 — Sacks, Shipping, Paper.
PPP-B-41 — Barrels; Wood, Slack.

Federal Standards:

Fed. Test Method Std. No. 158 — Cements, Hydraulic, Sampling, Inspection and Testing.

Fed. Std. No. 102 — Preservation, Packaging and Packing Levels.

Fed. Std. No. 123 — Marking for Domestic Shipment (Civilian agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material.—The natural cement covered by this specification shall be the product obtained by finely pulverizing calcined argillaceous limestone.

3.2 Chemical requirements.

3.2.1 *Ignition loss.* — The ignition loss of the cement shall not exceed 12 percent by weight.

3.2.2 *Insoluble residue.* — The insoluble residue of the cement shall be not less than 2 percent by weight.

3.3 Physical requirements. — Natural cement covered by this specification shall conform to the respective physical requirements in table I when tested in accordance with 4.1.

3.4 Additions during grinding. — Water and/or untreated calcium sulfate may be used. At the option of the manufacturer, nondeleterious materials other than water and/or calcium sulfate may be used as grinding aids in a total amount not in excess of 1 percent.

3.4.1 *Air-entraining addition.* — When air-entraining cement is specified, there shall be interground with the clinker an addition of such a nature and in such an amount that the resulting product complies with the applicable requirements of this specification. At the request of the purchaser, the manufacturer shall furnish a written statement giving the source, amount, and brand name of the addition used. At the request of the purchaser, the manufacturer shall also furnish test data (note) acceptable to the purchaser showing that air-entraining portland cement that contained this addition and met the applicable requirements of tables I and II of Federal Specification SS–C–192 had a durability factor of not less than 80 when tested in accordance with the procedures of Method 4201.

NOTE: Test data showing that an air-entraining addition complies with the requirements of ASTM Designation C–226 will be regarded as acceptable.

3.4.2 *Calcium chloride.* — When authorized by the purchaser, calcium chloride in an amount not to exceed 2 percent by weight of the natural cement may be interground with the cement to aid development of early strength of the concrete.

TABLE I.—*Physical requirements*

	Type	
	N	NA
Fineness, specific surface, sq.cm. per g:[1]		
Air permeability apparatus	---	---
Average value, minimum	6,000	6,000
Minimum value, any one sample	5,500	5,500
Soundness:[2]		
Autoclave expansion, maximum percent	0.50	0.50
Time of setting:[3]		
Gillmore test:		
Initial minimum minutes	60	60
Final, maximum hours	12	12
Air entrainment:[3]		
No agent used, maximum percent by volume	12	---
Using air-entraining agent, percent by volume	---	19±3
Compressive strength, minimum p.s.i.:[4]		
1 day in moist air, 6 days in water, p.s.i.	1,250	1,000
1 day in moist air, 27 days in water, p.s.i.	2,500	2,000

[1] Standard provisions as modified by 4.1.1 apply.

[2] The 1- × 1- × 10-inch bars shall remain firm and hard and show no signs of distortion, cracking, checking, pitting, or disintegration when subjected to autoclave expansion test. In addition, the standard provisions as modified by 4.1.2 and 4.1.3 apply.

[3] Standard provisions as modified by 4.1.2 apply.

[4] Standard provisions as modified by 4.1.2 and 4.2.1 apply.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling, inspection, and tests of the cement shall be in accordance with Federal Test Method Standard No. 158, except as modified herein.

4.1.1 The determination of fineness of the natural cement using the air-permeability apparatus shall be made at a porosity of 0.530 ± 0.005. The true specific gravity of the natural cement shall be used.

4.1.2 The cement used in preparing the test specimens for normal consistency, time of setting, autoclave expansion, strength and air-entrainment tests shall be a blend of 25 percent by weight of the natural cement being tested and 75 percent by weight of portland cement conforming to the requirements of Federal Specification SS–C–192, Cement, Portland, type II, unless otherwise specified by the purchaser. The natural cement and the portland cement shall be dry mixed to a uniform blend before water is added.

4.1.3 The autoclave expansion of the natural cement for the purpose of this specification shall be considered as the difference between the autoclave expansion of specimens prepared with the blend, as specified in 4.1.2, and that of specimens prepared with the portland cement as used in the blend.

4.2 **Acceptance and rejection.**

4.2.1 *Acceptance.* — The cement may be accepted on the basis of 7-day tests. Cement failing to meet the 7-day test for strength may be accepted if it meets the 28-day test requirement.

4.2.1.1 Cement failing to meet the initial test for autoclave soundness may be accepted if it passes a retest on a second and third series of test specimens.

4.2.2 *Rejection.* — The cement shall be rejected if it fails to meet any of the requirements of this specification subject to specific modification by the purchaser and/or to the modifications indicated herein.

4.2.2.1 Cement remaining in storage for a period longer than six months after test may be retested and shall be rejected if it fails to meet any of the requirements of this specification at that time.

4.2.2.2 Tentative or provisional acceptance of the cement shall not deprive the purchaser of the right of rejection in the event final tests or retests fail to meet applicable requirements.

5. PREPARATION FOR DELIVERY

5.1 For civil agency procurement, the levels of preservation, packaging and packing used are defined in Federal Standard No. 102.

5.2 **Packaging.**—The cement shall be delivered in bags, barrels, or in bulk, as specified. A bag shall contain 85 pounds net. A barrel shall contain 340 pounds net. All packages shall be in good condition at the time of the inspection. Packages varying more than 5 percent from the specified weight may be rejected; and if the average weight of packages in any shipment, as shown by weighing 50 packages taken at random, is less than that specified the entire shipment may be rejected. When shipping in bags, packing shall be in accordance with one of the following levels, as specified.

5.2.1 *Level A.* — Cement in quantities of 85 pounds net shall be packed in bags conforming to Federal Specification UU–S–48, X series.

5.2.2 *Level B.* — Cement in quantities of 85 pounds net shall be packed in bags conforming to Federal Specification UU–S–48 L/W series.

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5.2.3 *Level C.* — Cement shall be packed in a manner which will assure arrival at destination in satisfactory condition and which will be acceptable to the carrier at the lowest transportation cost.

5.3 Marking.

5.3.1 *Civil-agency procurement.* — Marking shall be in accordance with Federal Standard No. 123. When cement is delivered in packages, the name and brand of the manufacturer, the words "Natural Cement", the net weight and the absolute volume shall be plainly indicated thereon. Similar information shall be provided in the shipping advices accompanying the shipment of packaged or bulk cement.

5.3.2 *Military procurement.* — Marking shall be in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.** — Purchasers should specify the type of cement required and should exercise any desired options offered herein. (See 1.1, 4.1.2 and 5.1.)

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purposes, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 Federal Specifications do not include all types, classes, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, classes, etc., which are suitable for Federal Government requirements.

6.4 **Transportation description.** — The proper transportation description applicable to these items is—

Cement, natural or Portland
Carload weight 50,000 pounds.
Truckload weight 50,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Navy—Y.

Copies of this specification may be purchased for cents each.

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